

# A Nonexample for Continuous Inner Products in Directionally Euclidean Structures

Run fa\_banach\_001; arXiv:1005.2737

## Status

This packet records a likely valid partial result related to Talponen's question in [1, Section 2.1]: what kind of Banach spaces admit a continuous non-degenerate inner product? The source proves that  $\omega^*$ -separable dual is sufficient and notes that a continuous inner product yields a continuous linear injection into some  $c_0(\Gamma)$ . The result below gives a natural obstruction and shows that the latter  $c_0(\Gamma)$ -injection condition is not sufficient.

norming subspace  $Z \subset Y^*$  such that  $Y$  has DES in every direction  $f \in \mathbf{S}_Z$ .

**2.1. On boundedness of ellipsoids containing the unit ball.** In a Banach space a continuous non-degenerate inner product corresponds to a convex body that might be unbounded but does not contain a 1-dimensional linear subspace. There are plenty of such convex bodies, or inner products, according to the following fact:

**Proposition 2.5.** *Let  $X$  be a Banach space with a  $\omega^*$ -separable dual space. Then there exists an inner product  $(\cdot|\cdot)$  on  $X$  such that  $(x|x) \leq \|x\|^2$  and  $(x|x) > 0$  for  $x \in X$ ,  $x \neq 0$ . This in turn implies that there exists a continuous linear injection from  $X$  into  $c_0(\Gamma)$  for some set  $\Gamma$ .*

*Proof.* Let  $(x_n^*) \subset X^*$  be a sequence, which is  $\omega^*$ -dense in  $X^*$  and does not contain 0. It is easy to see that the mapping  $g: X \rightarrow \ell^\infty$  given by  $x \mapsto (\frac{x_n^*(x)}{\|x_n^*\|})$  is linear, contractive and injective. The latter property follows from the fact that  $(x_n^*)$  separates each  $x \in X$ ,  $x \neq 0$ , as it is  $\omega^*$ -dense. Next, observe that  $f: \ell^\infty \rightarrow \ell^2$  given by  $(z_n) \mapsto (2^{-n}z_n)$  is linear, contractive and injective. Now, the required inner product on  $X$  is induced by one in  $\ell^2$  via the composite mapping  $f \circ g$ .

In the latter claim we first form the completion  $H$  of  $(X, \|\cdot\|_2)$  and consider its orthonormal basis  $\{e_\gamma\}_{\gamma \in \Gamma}$ . The required map  $X \rightarrow c_0(\Gamma)$  is then given by  $x \mapsto \{(x|e_\gamma)\}_{\gamma \in \Gamma}$ .  $\square$

Spaces that admit a continuous linear injection into  $c_0(\Gamma)$  have been studied quite a bit (see e.g. [10]). We do not know exactly what kind of Banach spaces admit a continuous (non-degenerate) inner product. Note that a continuous inner product will induce a continuous norm on  $X$ , but according to the Open Mapping Principle this norm will be *complete* if and only if the original norm and the induced norm are equivalent.

Even though there are often continuous inner products on Banach spaces, the way the inner products are produced here by applying DES results in constructions that are sensitive to small changes of the norm, as it turns out. We also note that

## Source Question

In the source, a continuous non-degenerate inner product on a Banach space  $X$  is a positive definite bilinear form whose induced seminorm is bounded by the original norm. After Proposition 2.5, the source says that it is not known exactly which Banach spaces admit such an inner product.

## Proof Intuition

A continuous inner product on  $X$  is the same thing as a bounded linear injection from  $X$  into a Hilbert space: complete  $X$  in the Hilbert norm induced by the inner product. For  $X = c_0(\Gamma)$ , any bounded operator into a Hilbert space sends the coordinate unit vectors to a Hilbert family with uniformly bounded signed finite sums. Averaging over random signs turns that uniform bound into a square-summability estimate. Hence only countably many coordinate vectors can have nonzero image. If  $\Gamma$  is uncountable, injectivity is impossible.

## Main Result

**Lemma 1.** *Let  $\Gamma$  be any set,  $H$  a Hilbert space, and  $T : c_0(\Gamma) \rightarrow H$  a bounded linear operator. Then*

$$\{\gamma \in \Gamma : Te_\gamma \neq 0\}$$

*is countable, where  $e_\gamma$  denotes the  $\gamma$ -th unit vector.*

*Proof.* Put  $x_\gamma = Te_\gamma$ . For every finite  $F \subset \Gamma$  and every choice of signs  $\varepsilon_\gamma \in \{-1, 1\}$ , one has

$$\left\| \sum_{\gamma \in F} \varepsilon_\gamma x_\gamma \right\| = \left\| T \left( \sum_{\gamma \in F} \varepsilon_\gamma e_\gamma \right) \right\| \leq \|T\|,$$

because the signed coordinate sum has  $c_0(\Gamma)$ -norm one. Averaging the square of this inequality over independent random signs gives

$$\sum_{\gamma \in F} \|x_\gamma\|^2 = \mathbb{E} \left\| \sum_{\gamma \in F} \varepsilon_\gamma x_\gamma \right\|^2 \leq \|T\|^2.$$

Thus, for each  $n \in \mathbb{N}$ , the set  $\{\gamma : \|x_\gamma\| \geq 1/n\}$  is finite. The union of these finite sets over  $n$  contains every  $\gamma$  with  $x_\gamma \neq 0$ , so the support of the family  $(x_\gamma)$  is countable.  $\square$

**Theorem 2.** *If  $\Gamma$  is uncountable, then  $c_0(\Gamma)$  admits no continuous non-degenerate inner product.*

*Proof.* Assume first that  $B$  is a continuous non-degenerate inner product on  $c_0(\Gamma)$ . By continuity,  $B(x, x) \leq C\|x\|_\infty^2$  for some  $C < \infty$ . Let  $H$  be the Hilbert completion of  $c_0(\Gamma)$  under the norm  $x \mapsto B(x, x)^{1/2}$ , and let  $J : c_0(\Gamma) \rightarrow H$  be the canonical map. The displayed estimate makes  $J$  bounded, and non-degeneracy makes it injective.

This contradicts Lemma 1, since no bounded operator from  $c_0(\Gamma)$  into a Hilbert space can be injective when  $\Gamma$  is uncountable.  $\square$

**Corollary 3.** *The condition that a Banach space admit a continuous linear injection into some  $c_0(\Gamma)$  is not sufficient for the existence of a continuous non-degenerate inner product.*

*Proof.* For uncountable  $\Gamma$ , the identity map is a continuous injection  $c_0(\Gamma) \rightarrow c_0(\Gamma)$ , while Theorem 2 rules out a continuous non-degenerate inner product on  $c_0(\Gamma)$ .  $\square$

## Verification Notes

The proof uses only the standard random-sign identity in Hilbert spaces. No computational check is needed. A local duplicate search in the run indexes for the phrases “continuous inner product”, “Directionally Euclidean”, “inject Hilbert”, and “uncountable  $c_0$ ” found no existing packet for this obstruction.

## Limitations

This is a partial result. It does not classify all Banach spaces admitting continuous non-degenerate inner products. It also does not answer the source’s separate question whether DES in all or some directions passes to subspaces, ultrapowers, ultra-roots, or almost isometric copies.

## Human Review Recommendation

Check the equivalence between continuous inner products and bounded Hilbert injections, and the random-sign proof of Lemma 1. If both are accepted, this packet should be treated as a small but clean partial result for the source's continuous-inner-product classification question.

## References

- [1] J. Talponen, *Directionally Euclidean Structures of Banach Spaces*, arXiv:1005.2737.